

Kafka Streams - Introduction



1. What is Real-time Stream Processing?
2. What is Kafka?
3. Kafka Streams Architecture?

What is Stream Processing?



Data Streams are

Unbounded – No definite starting or ending
Often **Infinite** and **Ever Growing**
Sequence of data in small packets (KB)



Data Streams are

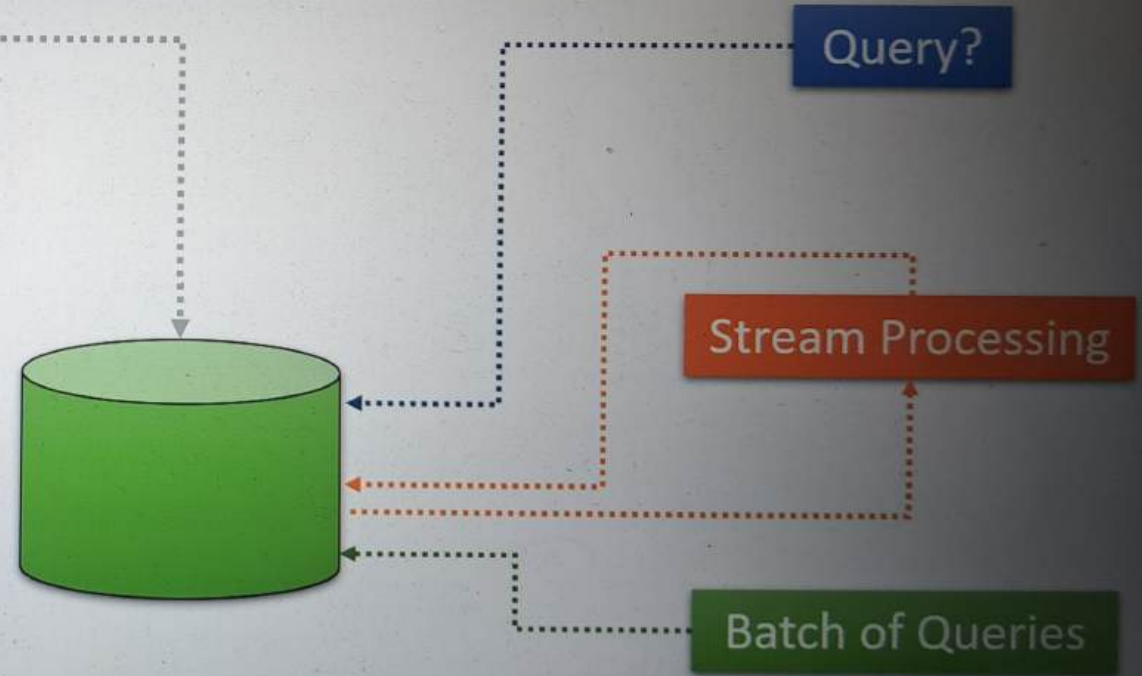
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Common Examples

1. Sensors
2. Log Entries
3. Click Streams
4. Transactions
5. Data Feeds

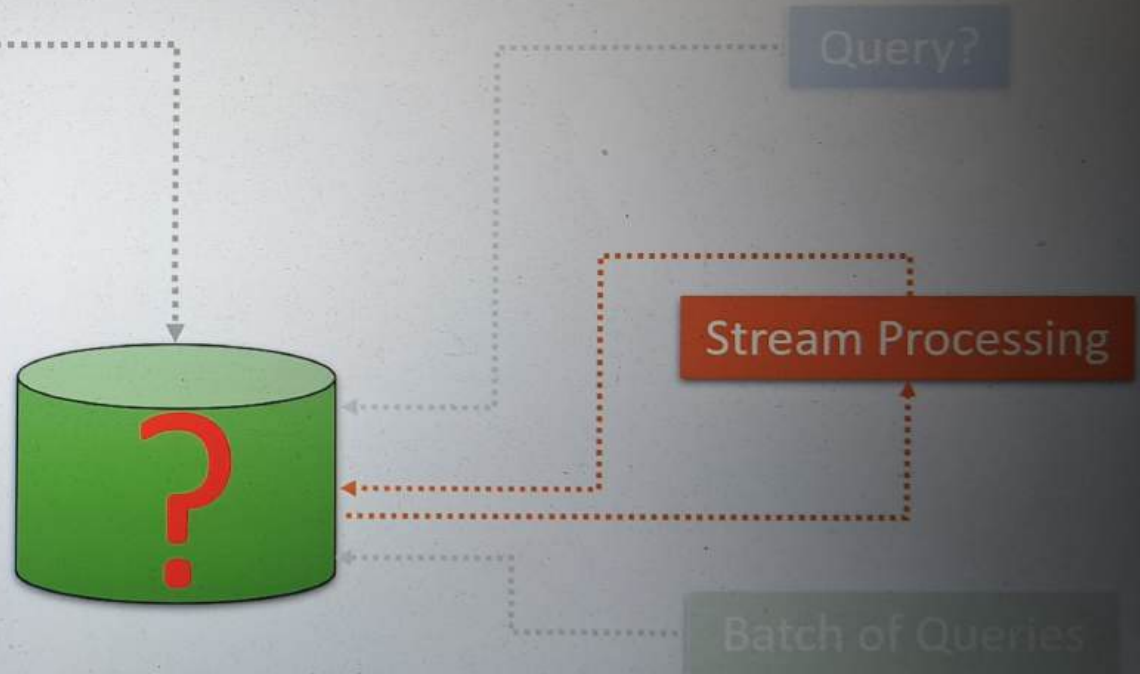
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Can we use

? Kafka Client APIs – Producer/Consumer
Kafka Connect



What is Kafka Streams?



What is Kafka Streams?

A Java/Scala Library....

Input data must be in Kafka Topic....

You can embed Kafka Streams in your microservices....

Deploy anywhere (No Cluster Needed)...

Out of the box parallel processing, scalability, and fault tolerance...

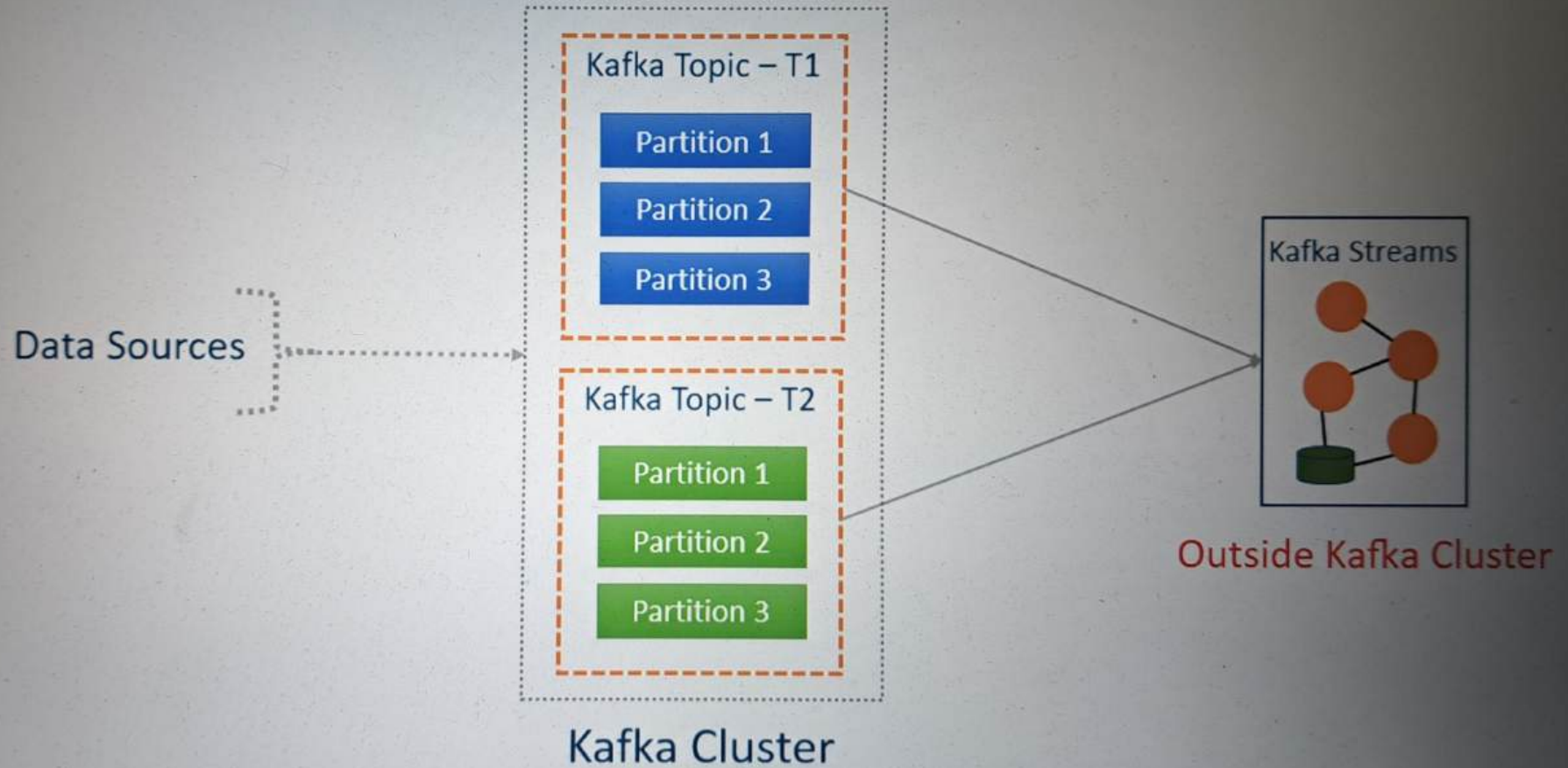


What Kafka Stream Offers?

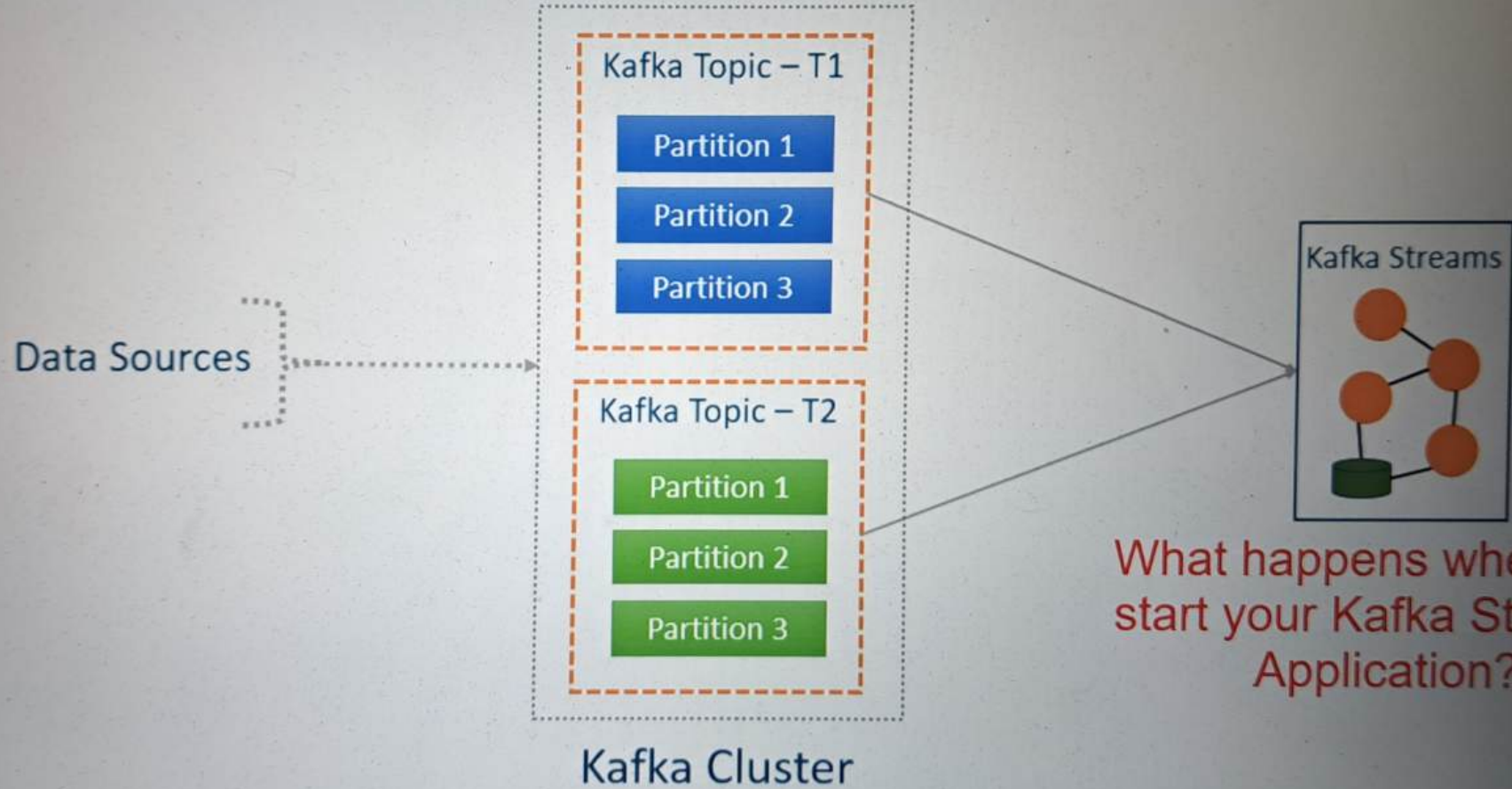
1. Working with streams/tables and interoperating with them
2. Grouping and Continuously updating Aggregates
3. Join streams, tables, and a combination of both
4. Create and manage fault-tolerant, efficient local state stores
5. Flexible Windowing capability
6. Flexible Time Schematics - Event time, Processing time, Latecomers, High watermark, Exactly-once processing, etc
7. Interactive Query - Serving other microservices
8. Unit testing tools
9. Easy to use DLS and extensibility to create custom processors
10. Inherent fault tolerance and dynamic scalability
11. Deploy in containers and manage using Kubernetes

Kafka Streams Architecture?

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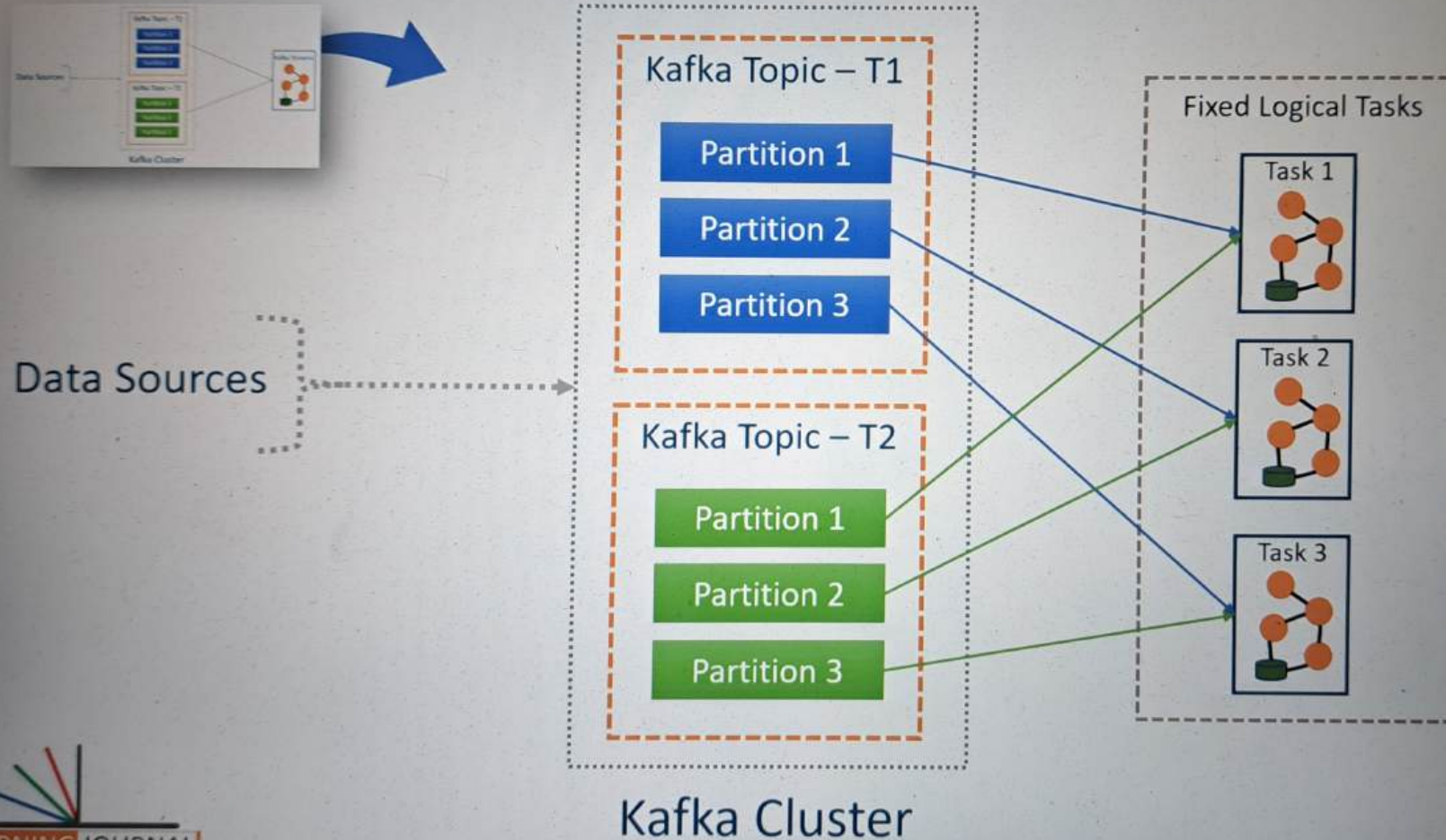


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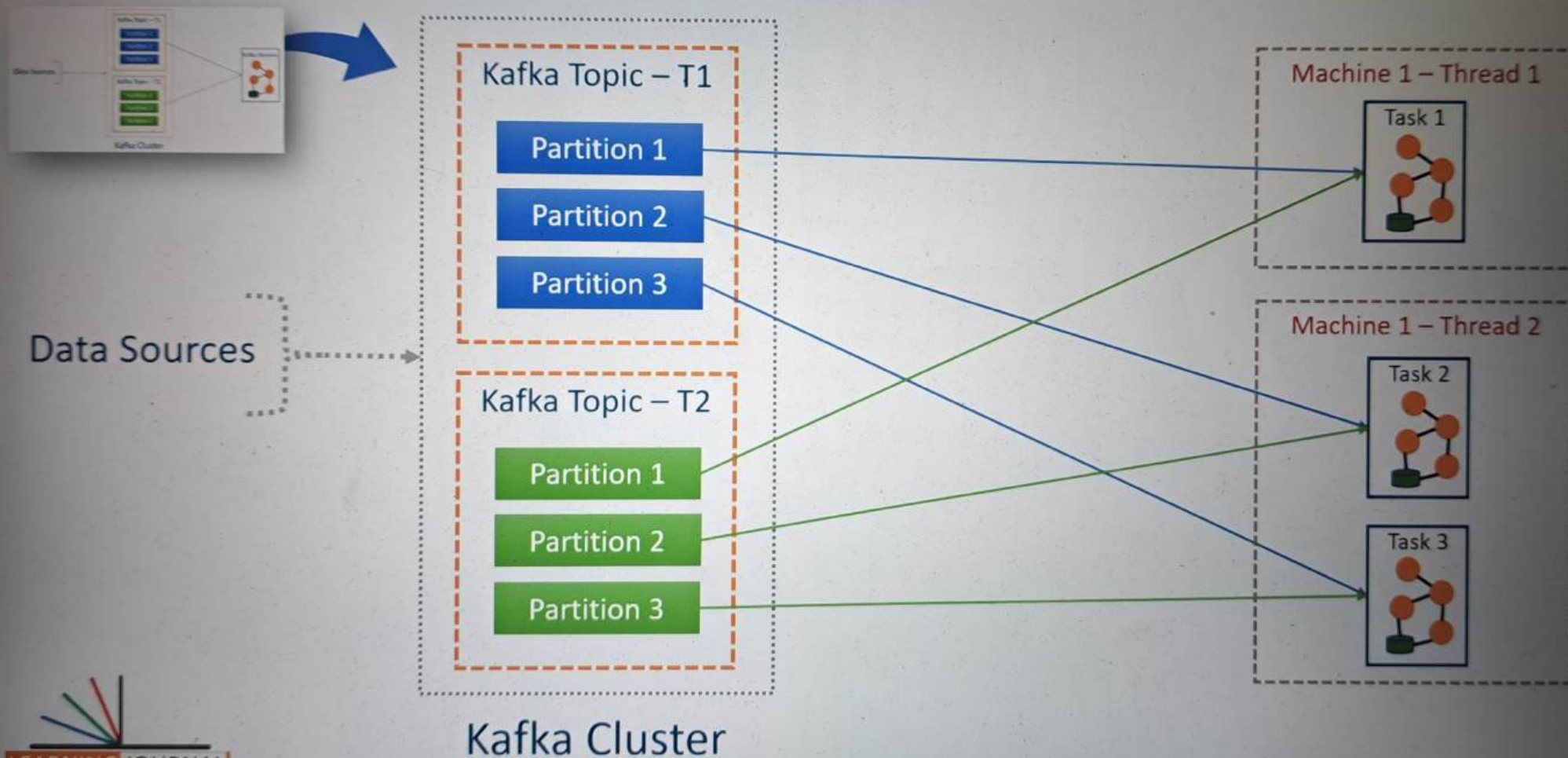


What happens when you start your Kafka Streams Application?

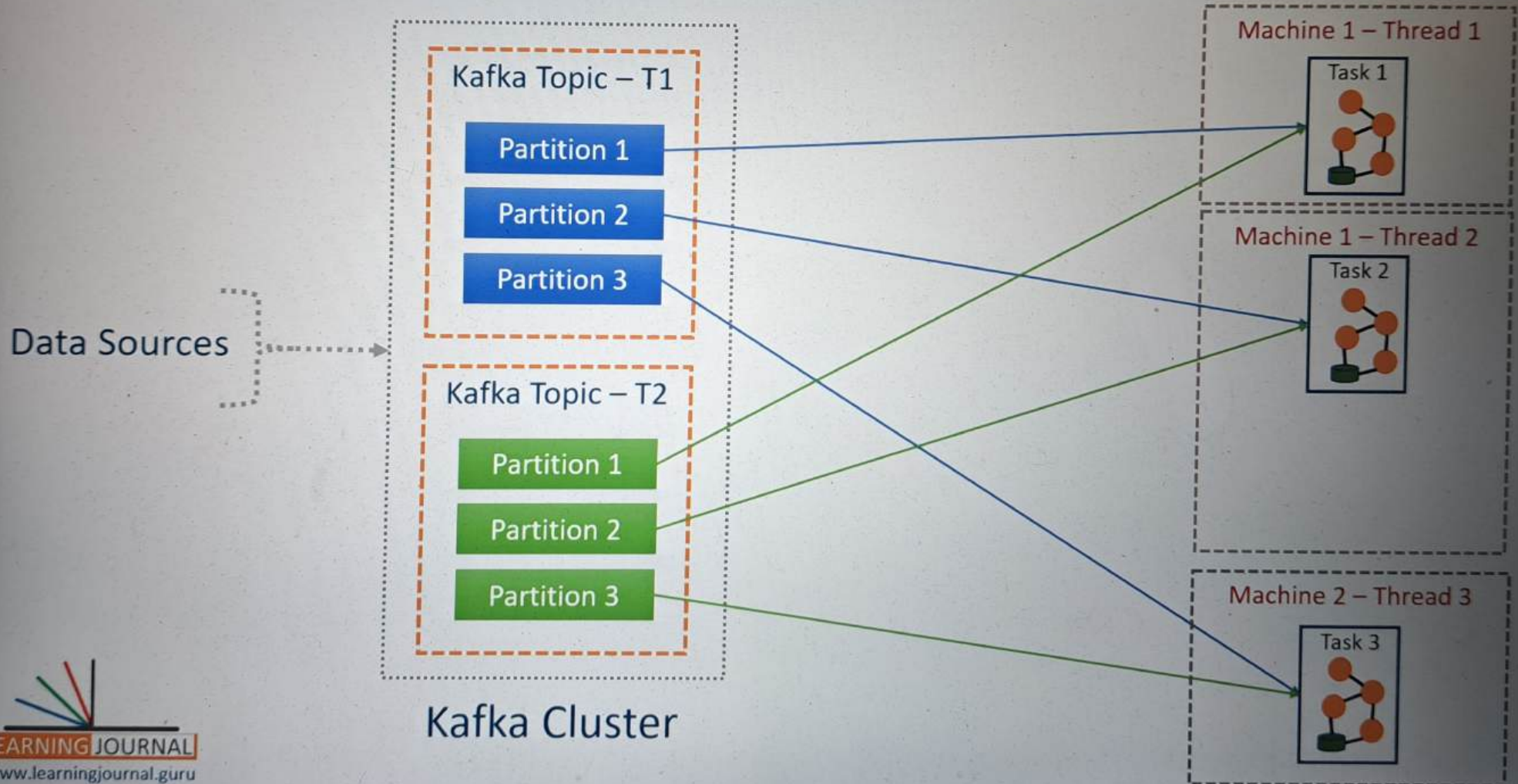
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